

Variables, functions, and control flow

Module 2 - Lesson 1

Overview

JavaScript programs are built from variables that store data, functions that package behavior, and control flow statements that decide what runs. This lesson covers the fundamentals you will use in every program.

Key Concepts

- `let` and `const` are the modern variable declarations; `const` is the default unless you need to reassign.
- Functions capture behavior and can take parameters and return values; arrow functions provide a concise syntax.
- Comparison operators and logical operators (`&&`, `||`, `!`) drive conditional logic.
- `if/else`, `switch`, and the ternary operator choose between code paths.
- Loops (`for`, `while`, `for...of`) repeat work; `map/filter/reduce` replace most hand-written loops.
- Scope determines where a variable is visible; prefer function and block scope over globals.

Example

```
const prices = [12, 45, 8];
const total = prices
  .filter(p => p > 10)
  .reduce((sum, p) => sum + p, 0);

function classify(score) {
  if (score >= 90) return 'A';
  if (score >= 80) return 'B';
  return 'C';
}
```

Summary

Variables hold state, functions package reusable behavior, and control flow directs execution. Mastering these fundamentals and the array methods built on them makes everything else in JavaScript easier.

Practice Checklist

- Rewrite a `for` loop using `map` and `filter` to transform an array.
- Write a function that validates a score and returns a grade using early returns.
- Explain the difference between `let`, `const`, and `var` and when each is appropriate.